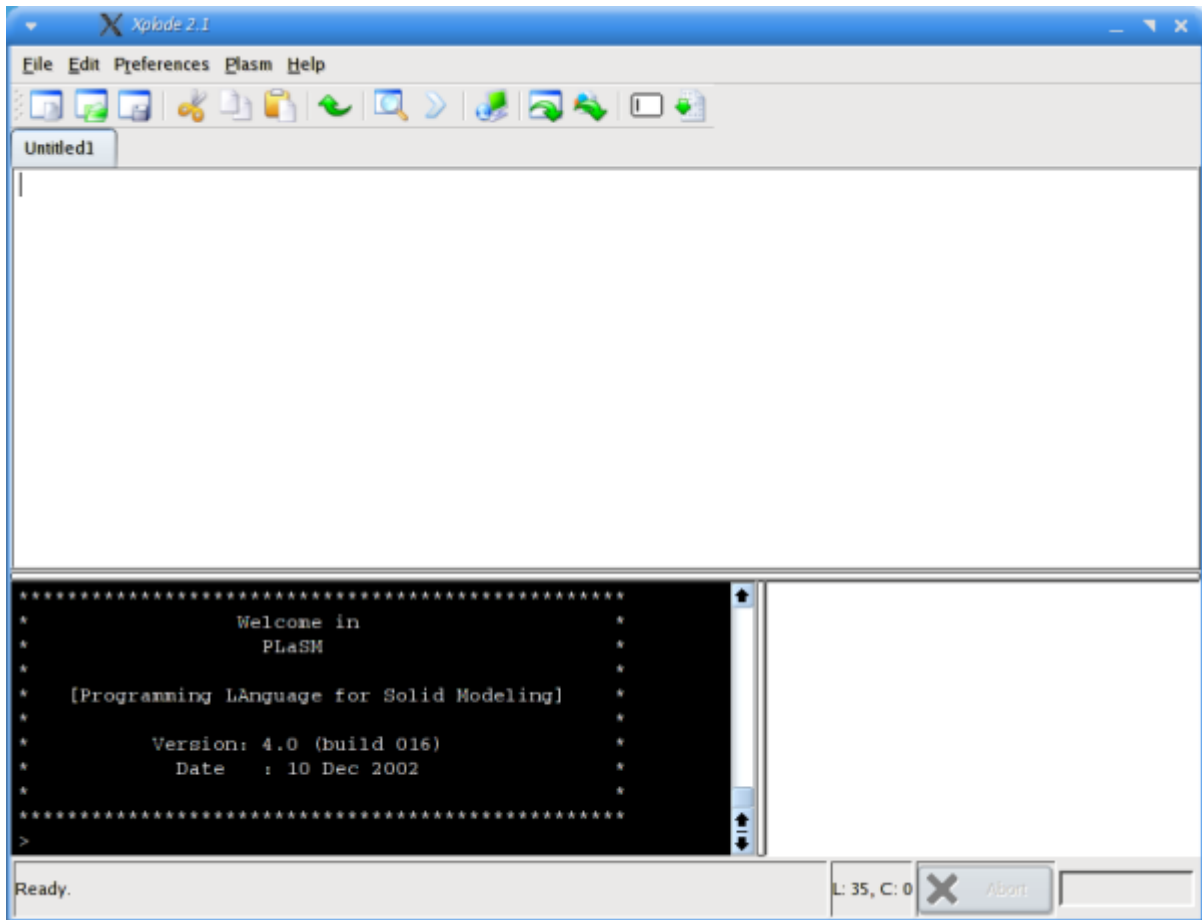


Readme

Installation files

with IDE:

If you are using Windows 7, Vista, or XP, the file **1320.exe** is recommended, as it includes an IDE (XPLode) that allows you to conveniently run PLaSM 4.0 programs (using the rightmost button in the IDE). After installation, navigate to the "plasm" folder within the "Program Files" directory and launch **Xplode.exe**, the PLaSM interpreter should then start automatically.



with Console:

Otherwise, you must use the file **windows-xp.msi** for Windows. This installs PLaSM 5.1 into the Program Files folder. After installation, open the PLaSM 5 folder within the Program Files directory and launch **plasm.exe**, a console window will open, allowing you to enter PLaSM 5 programs using the command (**plasm "..."**). The three dots represent the PLaSM programs.

[Picture]

Examples

$+: <4, 3>$

or

$[+, -]: <4, 3>$

or

$\text{CONS} : <\text{ID}, \text{SQRT}, \text{K:30}> : 2$

or

$\text{IF} : <\text{C:EQ:0}, \text{K:10}, \text{K:20}> : 0$

or

$\text{AA} : \text{SQRT} : <1, 2, 3, 4, 5>$

Lezione 7

Trasformazioni affini 3D
Traslazione e scalamento
Rotazione
Scorrimento

Rotazione intorno ad un asse qualunque

implementazione per trasformazione di coordinate

```
1 DEF Rotn (alpha::IsReal; n::IsVect) =
2   IF:< OR ~ [IsUp, IsZero] ~ s2, R:<1,2> ~ s1, Rot_n >:<
   alpha,n>
3 WHERE
4   Rot_n (alpha::IsReal; n::IsVect) = (MAT ~ TRANS):Q ~ R
   :<1,2>:alpha ~ MAT:Q,
5   Q = Mathom:<qx, qy, qz>,
6   qx = UnitVect:(<0,0,1> VectProd n),
7   qy = qz VectProd qx,
8   qz = UnitVect:n,
9   IsUp = AND ~ [C:EQ:0~s1, C:EQ:0 ~ s2, NOT ~ C:EQ:0 ~
   s3]
10 END;
11
12 DEF IsZero = AND ~ AA:(C:EQ:0);
```